

D430

# Fundamentals of Information Security

Armored Drill

8 concepts · 12 items · 4 transfer items

**What this is.** Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

**Source.** Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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# How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | _____  |                               |

# Items

1.

An unauthorized user changed dosage values in a hospital's pharmacy records. The system stayed online, and no data left the building. The hospital has 600 beds. Which CIA goal was violated?

- A. Confidentiality
- B. Availability
- C. Integrity
- D. All three, because any breach violates all three.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

2. [TRANSFER — reversed stem / different variable]

Redundant power supplies and tested backups mainly protect which CIA goal?

- A. Non-repudiation
- B. Availability
- C. Integrity
- D. Confidentiality

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

3.

A bank's login asks for a password and then the user's first pet's name. The bank recently won an award for its mobile app. Is this multifactor authentication?

- A. Yes, if the password is long enough.
- B. No; both factors are something you know.
- C. Yes; there are two separate steps.
- D. Yes; an award-winning app meets MFA standards.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

4. [TRANSFER — reversed stem / different variable]

A user logs in successfully but cannot open the payroll folder. Which step denied access?

- A. Identification
- B. Authorization
- C. Accounting
- D. Authentication

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

5.

At a 90-person nonprofit, the same accounts clerk can create new vendors and approve payments to them. The clerk has worked there eleven years with no complaints. Which principle is missing?

- A. Least privilege.
- B. No change needed, given the clerk's record.
- C. Separation of duties.
- D. Defense in depth.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

6.

A server worth \$200,000 would lose 25% of its value in a flood. Floods strike about once every 10 years. The server room was repainted last year. What is the annualized loss expectancy?

- A. \$500,000
- B. \$50,000
- C. \$20,000
- D. \$5,000

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

7. [TRANSFER — reversed stem / different variable]

An asset's ALE is \$12,000 and its SLE is \$48,000. What is the ARO?

- A. 0.25 (once every 4 years)
- B. \$36,000
- C. \$12,000
- D. 4

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

8.

A data center installs mantrap entrances, and the company also has an acceptable-use policy and firewall rules. The data center sits near an airport. What type of control is the mantrap?

- A. Technical, preventive.
- B. Physical, detective.
- C. Administrative, preventive.
- D. Physical, preventive.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

**9.** [TRANSFER — reversed stem / different variable]

A weekly review of access logs to spot unauthorized logins serves which control function?

- A.** Detective
- B.** Corrective
- C.** Deterrent
- D.** Preventive

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

**10.**

A law firm with offices in two cities relies on a single next-generation firewall, and its IT manager says it blocks 99% of threats. Which change best applies defense in depth?

- A.** Replace it with a larger firewall, since one strong control can cover everything.
- B.** Keep the setup, since 99% blocking is enough.
- C.** Buy a second identical firewall.
- D.** Add independent layers such as MFA, endpoint protection, segmentation, backups, and staff training.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

**11.**

A company needs partners to send it encrypted files without first sharing a secret key; the files average 2 GB and arrive weekly. Which approach fits?

- A.** Partners encrypt with the company's public key (usually wrapping a one-time symmetric key).
- B.** Hash each file with SHA-256.
- C.** Share one AES key with all partners by email.
- D.** Partners encrypt with the company's private key.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ___    |                               |

**12.**

A government agency labels files Secret or Top Secret, and users receive clearances. File owners cannot change the labels. The agency has 12 regional offices. Which access control model is this?

- A.** Role-based access control (RBAC).
- B.** Discretionary access control (DAC).
- C.** Mandatory access control (MAC).
- D.** Attribute-based access control only.

| Target word | The one fact that decides | Answer | [ ] sure [ ] unsure [ ] guess |
|-------------|---------------------------|--------|-------------------------------|
| _____       | _____                     | ____   |                               |

# STOP

Answers and dissection follow. Finish every item and every strip first.

# Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

## 1. Answer C Concept 1

An unauthorized user changed dosage values in a hospital's pharmacy records. The system stayed online, and no data left the building. ~~The hospital has 600 beds.~~ Which CIA goal was violated?

**Target word:** which goal failed **Decides:** Data was altered.

|   |                   |                                                              |
|---|-------------------|--------------------------------------------------------------|
| A | Confusion set     | No data was disclosed.                                       |
| B | Adjacent category | The system stayed online.                                    |
| C | <b>Correct</b>    | Unauthorized change to data is an integrity failure.         |
| D | Absolute language | Incidents usually hit one or two goals; here only integrity. |

## 2. Answer B Concept 1 · transfer

Redundant power supplies and tested backups mainly protect which CIA goal?

**Target word:** reverse: control to goal **Decides:** They keep systems running and restorable.

|   |                    |                                                                              |
|---|--------------------|------------------------------------------------------------------------------|
| A | Confusion set      | That is about proving who did an action.                                     |
| B | <b>Correct</b>     | They keep data and systems usable after failures.                            |
| C | Adjacent category  | Backups can help restore good data, but the main aim is uptime and recovery. |
| D | Inverted operation | Extra copies can add exposure; they do not protect secrecy.                  |

## 3. Answer B Concept 2

A bank's login asks for a password and then the user's first pet's name. ~~The bank recently won an award for its mobile app.~~ Is this multifactor authentication?

**Target word:** is it MFA **Decides:** Both are something you know.

|   |                   |                                                  |
|---|-------------------|--------------------------------------------------|
| A | Adjacent category | Password strength does not add a factor.         |
| B | <b>Correct</b>    | MFA needs different categories: know, have, are. |
| C | Confusion set     | Two steps of one factor type is not MFA.         |
| D | Camouflage        | Awards say nothing about factor types.           |

## 4. Answer B Concept 2 · transfer

A user logs in successfully but cannot open the payroll folder. Which step denied access?

**Target word:** reverse: symptom to AAA step **Decides:** Login worked; permission did not.

|   |                   |                                                       |
|---|-------------------|-------------------------------------------------------|
| A | Dropped element   | The user was already identified and verified.         |
| B | <b>Correct</b>    | Permissions decide what an authenticated user may do. |
| C | Adjacent category | Accounting records actions; it does not block them.   |
| D | Confusion set     | That step already succeeded.                          |

## 5. Answer C Concept 3

~~At a 90-person nonprofit, the same accounts clerk can create new vendors and approve payments to them. The clerk has worked there eleven years with no complaints.~~ Which principle is missing?

**Target word:** missing principle **Decides:** One person controls both halves of a sensitive task.

|   |                   |                                                             |
|---|-------------------|-------------------------------------------------------------|
| A | Adjacent category | Related, but the core gap is one person holding both steps. |
| B | Camouflage        | Controls should not rest on trust in one person.            |
| C | <b>Correct</b>    | Creating and approving should be split between people.      |
| D | Confusion set     | About layered technical defenses, not splitting a task.     |

## 6. Answer D Concept 4

A server worth \$200,000 would lose 25% of its value in a flood. Floods strike about once every 10 years. ~~The server room was repainted last year.~~ What is the annualized loss expectancy?

**Target word:** ALE **Decides:**  $SLE = 200,000 \times 0.25$ ;  $ARO = 0.1$ .

|   |                    |                                                   |
|---|--------------------|---------------------------------------------------|
| A | Inverted operation | Multiplies by 10 years instead of 0.1 per year.   |
| B | Adjacent category  | That is the SLE, a per-event loss.                |
| C | Dropped element    | $\$200,000 \times 0.1$ skips the exposure factor. |
| D | <b>Correct</b>     | $SLE \$50,000 \times ARO 0.1 = \$5,000$ .         |

## 7. Answer A Concept 4 · transfer

An asset's ALE is \$12,000 and its SLE is \$48,000. What is the ARO?

**Target word:** solve for ARO **Decides:**  $ARO = ALE / SLE$ .

|   |                    |                                |
|---|--------------------|--------------------------------|
| A | <b>Correct</b>     | $12,000 / 48,000 = 0.25$ .     |
| B | Confusion set      | Subtracts instead of dividing. |
| C | Wrong stem number  | The ALE from the stem.         |
| D | Inverted operation | $SLE / ALE$ flips the ratio.   |

## 8. Answer D Concept 5

A data center installs mantrap entrances, ~~and the company also has an acceptable-use policy and firewall rules. The data center sits near an airport.~~ What type of control is the mantrap?

**Target word:** category and function **Decides:** A mantrap physically stops entry.

|   |                    |                                                          |
|---|--------------------|----------------------------------------------------------|
| A | Adjacent category  | That fits the firewall rules.                            |
| B | Inverted operation | It blocks entry rather than detecting it after the fact. |
| C | Confusion set      | That fits the acceptable-use policy.                     |
| D | <b>Correct</b>     | It is a physical barrier that stops unauthorized entry.  |

## 9. Answer A Concept 5 · transfer

A weekly review of access logs to spot unauthorized logins serves which control function?

**Target word:** function **Decides:** It finds problems after they happen.

|   |                    |                                                         |
|---|--------------------|---------------------------------------------------------|
| A | <b>Correct</b>     | Log review discovers events that already occurred.      |
| B | Adjacent category  | Correction comes after detection.                       |
| C | Confusion set      | Deterrents discourage attempts; this review finds them. |
| D | Inverted operation | Reviewing logs does not stop the login.                 |

## 10. Answer D Concept 6

A law firm ~~with offices in two cities~~ relies on a single next-generation firewall, ~~and its IT manager says it blocks 99% of threats~~. Which change best applies defense in depth?

**Target word:** layered protection **Decides:** One control is a single point of failure.

|   |                   |                                                     |
|---|-------------------|-----------------------------------------------------|
| A | Absolute language | No single control stops every threat.               |
| B | Camouflage        | The remaining 1% still reaches unprotected systems. |
| C | Confusion set     | Redundancy for one layer is not a new layer.        |
| D | <b>Correct</b>    | Different layers catch what others miss.            |

## 11. Answer A Concept 7

A company needs partners to send it encrypted files without first sharing a secret key; ~~the files average 2 GB and arrive weekly~~. Which approach fits?

**Target word:** encryption without a shared secret **Decides:** No prior shared key.

|   |                    |                                                                             |
|---|--------------------|-----------------------------------------------------------------------------|
| A | <b>Correct</b>     | Only the company's private key can decrypt; no secret is shared in advance. |
| B | Confusion set      | Hashing checks integrity; it does not hide content.                         |
| C | Dropped element    | Needs the shared secret the company wants to avoid, sent insecurely.        |
| D | Inverted operation | The private key is never shared.                                            |

## 12. Answer C Concept 8

A government agency labels files Secret or Top Secret, and users receive clearances. File owners cannot change the labels. ~~The agency has 12 regional offices~~. Which access control model is this?

**Target word:** model **Decides:** System-enforced labels and clearances.

|   |                    |                                                                                     |
|---|--------------------|-------------------------------------------------------------------------------------|
| A | Adjacent category  | RBAC ties permissions to job roles, not clearance labels.                           |
| B | Inverted operation | In DAC, owners decide access.                                                       |
| C | <b>Correct</b>     | The system enforces labels that owners cannot override.                             |
| D | Confusion set      | ABAC uses policy rules over many attributes; classic labels and clearances are MAC. |

## Triage grid

Write each item number in the box it belongs to.

|                       | Right                                                 | Wrong                                                                                 |
|-----------------------|-------------------------------------------------------|---------------------------------------------------------------------------------------|
| <b>Sure</b>           | Owned.                                                | <b>Concept inversion — top priority.</b> A wrong mental model; it survives rewording. |
| <b>Unsure / guess</b> | Recognition, not mastery. Re-drill with the transfer. | Not learned yet.                                                                      |

## Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

| Trap               | What it does                                                 | In this drill | Your misses |
|--------------------|--------------------------------------------------------------|---------------|-------------|
| Dropped element    | Leaves out a required term or step.                          | 3             |             |
| Wrong stem number  | A figure copied from the prompt that is not the answer.      | 1             |             |
| Adjacent category  | The right idea in the wrong bucket.                          | 9             |             |
| Inverted operation | Direction flipped: add/subtract, cause/effect, higher/lower. | 7             |             |
| Absolute language  | Always, never, guarantee — where judgment applies.           | 2             |             |
| Confusion set      | Every option is real; the test is exact role-matching.       | 11            |             |
| Camouflage         | Answers a stem fact that was never the question.             | 3             |             |

# Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

| # | Concept                                         | Rule                                                                                                                                                                           | Trap                                                                                       | Tell                                                      |
|---|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|-----------------------------------------------------------|
| 1 | <b>The CIA triad</b>                            | Confidentiality = only authorized people see data; integrity = data is not altered improperly; availability = systems and data are there when needed.                          | Calling every breach a confidentiality problem.                                            | What went wrong: exposed, changed, or unreachable.        |
| 2 | <b>Authentication and AAA</b>                   | Identification claims who you are, authentication proves it, authorization sets what you may do, accounting records what you did; MFA needs factors from different categories. | Counting two passwords or questions as MFA, or mixing up authentication and authorization. | Two login steps, or “logged in but cannot open.”          |
| 3 | <b>Least privilege and separation of duties</b> | Least privilege limits each person to what the job needs; separation of duties splits a sensitive task so no one person controls it end to end.                                | Treating the two as the same, or trusting tenure instead of controls.                      | One person can both create and approve.                   |
| 4 | <b>Quantitative risk (ALE)</b>                  | $SLE = \text{asset value} \times \text{exposure factor}$ ; $ALE = SLE \times ARO$ .                                                                                            | Stopping at SLE or skipping the exposure factor.                                           | Asset value, percent loss, and a frequency.               |
| 5 | <b>Control categories and functions</b>         | Categories: administrative, technical, physical. Functions: preventive, detective, corrective, deterrent.                                                                      | Mixing category with function, or labeling a lock as technical.                            | A named control and “what type.”                          |
| 6 | <b>Defense in depth</b>                         | Use several independent layers so one failure does not expose the asset.                                                                                                       | Relying on one strong control.                                                             | A single product described as blocking almost everything. |
| 7 | <b>Encryption and hashing</b>                   | Symmetric uses one shared key; asymmetric encrypts with the recipient’s public key and decrypts with their private key; hashing checks integrity and cannot be reversed.       | Encrypting with a private key for secrecy, or calling hashing encryption.                  | “Without sharing a secret key.”                           |
| 8 | <b>Access control models</b>                    | DAC: owners decide. MAC: system-enforced labels and clearances. RBAC: permissions follow job roles.                                                                            | Calling any label-based system RBAC.                                                       | Labels and clearances vs. owners vs. roles.               |

**Course scope (public description):** This course lays the foundation for understanding terminology, principles, processes, and best practices of information security at local and global levels. It further provides an overview of basic security vulnerabilities and countermeasures for protecting information assets through planning and administrative controls within an organization. This course has no prerequisites.